

Assessment - 02

1. $45 \text{ MHz} \rightarrow 45000 \text{ KHz}$

$$\frac{45000}{25} \Rightarrow 1800 \text{ KHz}$$

Full duplex require 2 Simplex channel

$$\Rightarrow \frac{1800}{2} \Rightarrow 900$$

2. 560 total voice channel

7-cell frequency reuse traffic

$$\Rightarrow \frac{560}{7} \Rightarrow 80 \text{ channel}$$

3. 100 GSM carriers, 8 time slot

$$\therefore \text{total capacity} \Rightarrow 100 \times 8 \Rightarrow 800$$

4. Power Ratio

$$\Rightarrow 10 \log_{10} (80/4)$$

$$\Rightarrow 80/4 \Rightarrow 20$$

$$\Rightarrow 10 \log_{10} (20) \rightarrow 13 \text{ dB}$$

5.

i) -60 dBm

ii) -92 dBm

$$\Rightarrow -60 - (-92)$$

$$\Rightarrow 32 \text{ dBm}$$

6.

$$\Rightarrow 42 \text{ MHz} \div 300 \text{ KHz}$$

$$\Rightarrow 140 \text{ channels}$$

$$\Rightarrow 140 \div 7$$

$$\Rightarrow 20 \text{ channels}$$

7.

Reduces Power $= 2^4 \Rightarrow 16$

$$\Rightarrow \frac{64}{16} \Rightarrow 4W$$

8.

Inverse - Square law ($n=2$)

$$\Rightarrow 2^2 \Rightarrow 4 \times \text{more transmit}$$

9.

cluster contains $\Rightarrow 4$ cells

each is voice channel

$$\therefore 4 \times 15 \Rightarrow 60 \text{ channels}$$

10.

Converting 100 km/h to 27.78 m/s

using $f_c(\text{Hz}) = \frac{v}{\lambda}$

$$= \left(\frac{27.78}{3 \times 10^8} \right) \times 900 \times 10^6$$

$$\Rightarrow 83 \text{ Hz}$$

11.

Frame duration is 5 ms

8 equal time slots

$$\Rightarrow 5 \div 8$$

$$\Rightarrow 0.625 \text{ ms}$$

12.

$$24 \text{ MHz} \Rightarrow 24000 \text{ KHz}$$

$$\frac{24000}{200} \Rightarrow 120 \text{ carriers.}$$

13.

$$8W = 39 \text{ dB}$$

$$39 - 33 \Rightarrow 6 \text{ dB}$$

14)

$$FSPL = 32.44 + 20 \log_{10} (1 \text{ MHz}) + 20 \log_{10} (2.5)$$

$$\rightarrow 1900$$

$$FSPL = 32.44 + 20 \log_{10} (1900) + 20 \log_{10} (2.5)$$

$$\Rightarrow 106 \text{ dB}$$

15)

250 call x 6 cells

$$\Rightarrow 1500 \text{ calls}$$

16)

$$3 \text{ min} = 180 \text{ sec}$$

$$\therefore 4000 \times 180 \Rightarrow 720,000 \text{ active calls}$$

17)

- each of 150 channels occupies 25 KHz

- Total bandwidth

$$= 150 \times 25$$

$$= 3750 \text{ KHz} \Rightarrow 3.75 \text{ MHz}$$

18)

30 guard bands x 40 KHz

$$\Rightarrow 1200 \text{ KHz}$$

$$\Rightarrow 1.2 \text{ MHz}$$

19)

30 handoff per hour

cell diameter 2 x 15 km

$$\Rightarrow 90 \div 3 \Rightarrow 30$$

20)

$$8 \times 40 \Rightarrow 320$$

$$320 \times 250$$

$$\Rightarrow 80,000 \text{ users}$$

21)

$$\text{cell area} = R^2 \Rightarrow 0.4^2 \Rightarrow 0.16 \text{ Km}^2$$

$$\frac{400}{0.16} \Rightarrow 2500 \text{ cells}$$

22)

$$6 \text{ MHz} = 6000 \text{ KHz}$$

$$\frac{6000}{30} \Rightarrow 200 \text{ channels}$$

23)

$$T_{\text{traffic}} = 1200 \times 4 \times (2.5/60)$$

$$\Rightarrow 200 \text{ Erlangs}$$

24)

$$40 \text{ MHz} = 40,000 \text{ KHz}$$

$$\frac{40,000}{400} \Rightarrow 100 \text{ users}$$

25)

$$\therefore 1200 \times 100 \Rightarrow 120,000 \text{ users}$$

26)

$$200 \text{ cells}, 4 \text{ cells per round}$$

$$\Rightarrow \frac{200}{4} \Rightarrow 50 \text{ rounds}$$

27)

$$40,000 \times 3 \Rightarrow 120,000 \text{ connections}$$

$$\text{capacity is } 100,000$$

$$\text{Blocking} = 20000 \div 120,000 \times 100$$

$$\Rightarrow 16.6\%$$

28)

$$\Rightarrow 600,000 \text{ of } 15\%$$

$$\Rightarrow 90,000 \text{ Subscribers}$$

29)

$$\Rightarrow (9000 \div 12,000) \times 100$$

$$\Rightarrow 75\%$$

$$30) \quad = 48 \text{ MHz} \div 200 \text{ kHz} \\ = 240, \text{ reuse factor} = 4 \\ \Rightarrow 240 / 4 \Rightarrow 60 \text{ channels}$$

$$31) \quad \Rightarrow 30 \text{ MHz} \div 200 \text{ kHz} \\ \Rightarrow 150 \text{ channel}$$

$$32) \quad = 991 \div 2500 \\ = 2475$$

$$33) \quad 12 \text{ W} = 40.8 \text{ dBm}$$

$$\text{system loss} = 4 \text{ dB}$$

$$\Rightarrow 40.8 - 45$$

$$\Rightarrow -4 \text{ dBm}$$

$$34) \quad = 7 \times 900 \\ \Rightarrow 63000 \text{ users}$$

$$85) \quad \frac{4.8 \text{ ms}}{8} \Rightarrow 0.6 \text{ ms per slot}$$

$$36) \quad = 150 \text{ km/h} = 41.67 \text{ ms}$$

$$= (41.67 / 3 \times 10^8) \times 1.8 \times 10^9$$

$$\Rightarrow 250 \text{ Hz}$$

$$87) \quad \Rightarrow 70000 \times 4 = 280,000 \text{ calls}$$

$$\Rightarrow \text{while capacity } 250,000$$

$$\Rightarrow 250000 \div 280000 \times 100 \\ \Rightarrow 10.7\%$$

$$38) \quad 9 \times 600$$

$$\Rightarrow 5400 \text{ users}$$

39) out of 60,000, 0.4 %
 $\Rightarrow 240 \text{ cal / Sec}$

40) $43 \text{ dBm} - (-87 \text{ dBm})$
 $\Rightarrow 130 \text{ dBm}$

41) $\Rightarrow 150 \times 2.5$
 $\Rightarrow 375 \text{ Km}^2$

42) Total channel = $48 \text{ MHz} = 280 \text{ KHz}$
 $= 240$
 with reuse factor = 6
 $\Rightarrow 240 \div 6 = 40$
 $\Rightarrow 40 \text{ channel}$

43) $\Rightarrow \frac{150,000 \text{ users}}{6,000 \text{ Paging cycle}}$
 $\Rightarrow 25 \text{ cycles}$

44) call run = 9
 min = 3
 $\Rightarrow \frac{9}{3} \Rightarrow 3$

45) signal = -85 dBm
 noise floor = -115 dBm
 $-85 - (-115)$
 $\Rightarrow 30 \text{ dBm}$

46) 25 users $\times 150 \text{ Kbps}$
 $\Rightarrow 37500 \text{ Kbps}$
 $\Rightarrow 37.5 \text{ Mbps}$

47)

- 1500 Subscribers
- 4 calls in hours
- 60 min
- 2.5 min

$$= 1500 \times 4 \times (2.5 / 60)$$

$$= 250 \text{ Erlangs}$$

48) $= 250 \times 12 \text{ KHz}$

- = 3000 KHz
- = 3 MHz

49)

$$= 1500 \times 600$$

$$= 900000 \text{ Kbbs}$$

$$\Rightarrow 900 \text{ MBbs}$$

50)

Coverage per tower $\rightarrow \pi \times 4^2$

$$\rightarrow 50.27 \text{ km}^2$$

$$\Rightarrow 1200 \times 0.00350 \div 50.27$$

$$\Rightarrow 6.96 \approx 7$$